

Hailey Petsch-Sinclair

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PROFESSIONAL STATEMENT

Highly motivated and results-driven Mechanical Engineer and strategist with an understanding of R&D testing, design, manufacturing, and quality with experience in product creation, business case development, and AI transformation. With a track record of successfully managing large-scale projects, I am committed to leveraging knowledge developed from a multi-faciated career to drive organizational improvement and development.

EDUCATION

Arizona State University B.S. Mechanical Engineering	GPA: 3.6 (Distinguished Graduate)	08/2020-05/2024
Louisiana State University M.B.A. International Business		03/2026-07/2027

WORK AND ACADEMIC EXPERIENCE

Product Strategy Sr. Analyst Engineer 05/2026-Present
Honda Research and Development Center

- Spearheaded product ideation and concept development for next-generation off-road vehicle platforms, aligning feature roadmaps with emerging enthusiast market segments and competitive whitespace.
- Executed comprehensive scenario planning and market simulations to evaluate long-term powertrain and vehicle trends, future-proofing product portfolios against regulatory and economic shifts.
- Authored rigorous business cases and financial feasibility models for new vehicle variants, securing executive alignment and cross-functional resource allocation.
- Synthesized Voice of Customer data, ethnographic research, and behavioral trends to define product requirements and target customer personas for future mobility solutions.
- Championed AI transformation initiatives as a designated "Digital Top Gun," driving the adoption of advanced generative and predictive analytics tools across product planning workflows.
- Pioneered AI-assisted digital storytelling and immersive visualization frameworks to communicate complex product vision and strategic business cases to executive leadership.
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Vehicle Dynamics Test Engineer 05/2023-08/2023,05/2024-Present
Honda Research and Development Center

- Researched consumer trends and automotive market to propose business case to help expand vehicle competitiveness, increase sales volume, and increase engineering capability to existing vehicle platform.
- Pioneered and led a cross-functional project consisting of 40 members focused on improving customer access to vehicle capability through the application of off road systems. Utilized

technical knowledge of vehicle dynamics to analyze performance metrics, identify improvement opportunities, and implement suspension modifications to optimize customer satisfaction.

- Benchmarked competitor vehicles to identify gaps in performance and used data to create competitive vehicles.
- Analyzed data for market quality concerns and ensured product quality and safety by supporting suspension-related investigations and identifying root causes.

Engine Durability Test Engineering intern
Honda Research and Development Center

05/2022-08/2022

- Performed engine durability testing (dynamometer and vehicle) using measurement techniques involving accelerometers, strain gauges, and thermocouples to verify engine durability lifespan.
- Analyzed market quality concerns by performing targeted testing and proposing effective countermeasures based on data analysis.
- Collaborated with a global engineering team, effectively navigating language and cultural differences to achieve project goals.

PROFESSIONAL SKILLS AND ACCOMPLISHMENTS

- Fisher College of Business: Mastering Leadership and Negotiation for Effective Change Management certificate
- UCLA Agentic Ai certificate
- UCLA Ai marketing certificate
- Proficiency with Microsoft Office
- Agile and flexible to learn and take on new activity